



Mycosynthesis of silver and gold nanoparticles: Optimization, characterization and antimicrobial activity against human pathogens



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ARTICLE INFO

Article history:

Received 7 July 2015

Received in revised form

14 September 2015

Accepted 28 September 2015

Available online 3 October 2015

Keywords:

Soil fungi

Aspergillus terreus

Silver nanoparticles

Gold nanoparticles

Antimicrobial activity

ABSTRACT

This study was aimed to isolate soil fungi from Kolli and Yercaud Hills, South India with the ultimate objective of producing antimicrobial nanoparticles. Among 65 fungi tested, the isolate, Bios PTK 6 extracellularly synthesized both silver and gold nanoparticles with good monodispersity. Under optimized reaction conditions, the strain Bios PTK 6 identified as *Aspergillus terreus* has produced extremely stable nanoparticles within 12 h. These nanoparticles were characterized by UV–vis. spectrophotometer, HR-TEM, FTIR, XRD, EDX, SAED, ICP-AES and Zetasizer analyses. *A. terreus* synthesized 8–20 nm sized, spherical shaped silver nanoparticles whereas gold nanoparticles showed many interesting morphologies with a size of 10–50 nm. The presence and binding of proteins with nanoparticles was confirmed by FTIR study. Interestingly, the myco derived silver nanoparticles exhibited superior antimicrobial activity than the standard antibiotic, streptomycin except against *Staphylococcus aureus* and *Bacillus subtilis*. The leakage of intracellular components such as protein and nucleic acid demonstrated that silver nanoparticles damage the bacterial cells by formation of pores, which affects membrane permeability and finally leads to cell death. Further, presence of nanoparticles in the bacterial membrane and the breakage of cell wall were also observed using SEM. Thus, the obtained results clearly reveal that these antimicrobial nanoparticles could be explored as promising candidates for a variety of biomedical and pharmaceutical applications.

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1. Introduction

Decreasing the size of nanoparticles has pronounced effect on the physical properties that are significantly different from their corresponding parent materials. The strong relationship between size and properties of nanoparticles has offered countless opportunities for many scientific breakthroughs. The extremely surprising activities of nanoparticles have enormous potential for modern technology emphasizing their use in human wellbeing (Heiligt and Niederberger 2013; Mullai et al., 2013). In recent times, the green chemistry procedure which utilizes microorganisms and plants for nanoparticles preparation has turned as a viable alternative to conventional physicochemical methods since it is facile, rapid, cost-effective, and environmentally benign. Use of microorganisms as cell factories for producing nanoparticles has received much scientific attention over the past decade. Recently, the

microbe mediated synthesis method has emerged as a burgeoning area of research in the field of nanobiotechnology (Narayanan and Sakthivel, 2010).

Currently, an exhaustive study on biological synthesis of nanoparticles has been carried out using a wide array of microorganisms such as algae, bacteria, actinomycetes, fungi, yeasts, and viruses (Narayanan and Sakthivel, 2010; Thakkar et al., 2010). Among them, fungi are more advantageous because the fungal mycelial mesh can withstand flow pressure, agitation and other conditions in bioreactors or other chambers compared to plant materials and bacteria. They are fastidious to grow, easy to handle and synthesize nanoparticles (Gade et al., 2008; Thakkar et al., 2010). The fungal mediated extracellular synthesis method has attracted a great deal of interest owing to its simplicity, no further downstream processing, and lesser time consumption in contrast to intracellular synthesis (Mishra et al., 2011). In addition, the size and shape of extracellularly synthesized nanoparticles can also be manipulated by controlling pH, temperature, substrate concentration (metal ions), and reaction time (Krishnaraj et al., 2012; Sathishkumar et al., 2010). Although nanoparticles with controlled

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size and shape could be achieved through intracellular synthesis, product harvesting and recovery are laborious and expensive (Fayaz et al., 2010; Thakkar et al., 2010). Therefore, the extracellular synthesis methods could be ideally used for large scale production of nanoparticles for several industrial applications (Mishra et al., 2011). Importantly, fungi have also secreted fairly large amount of proteins and secondary metabolites extracellularly and hence, the fungal biomass could reduce the metal ions more easily leading to the rapid formation of nanoparticles (Gade et al., 2008). Because of these advances over other methods, the myco-based extracellular synthesis method is often considered a better resource for higher productivity of nanoparticles (Du et al., 2011; Muhsin and Hachim, 2014).

Many researchers have reported the extracellular synthesis of both silver and gold nanoparticles using a variety of fungi. In *Fusarium oxysporum*, the reduction of Ag^+ ions has occurred through the release of reductase enzymes (Ahmad et al., 2003). Silver nanoparticles have rapidly been produced using a marine fungus, *Penicillium fellutanum* isolated from coastal mangrove sediment of South India (Kathiresan et al., 2009). A fast growing and non-pathogenic fungus, *Trichoderma viride* has been used for silver nanoparticles preparation (Fayaz et al., 2010). Du et al. (2011) have reported that the culture supernatant of *Penicillium* species supported the extracellular production of gold nanoparticles within a minute. Interestingly, a few researchers have used isolated fungal components for extracellular production of nano-sized silver and gold particles. For instance, Apte et al. (2013) have prepared silver and gold nanostructured materials by employing L-DOPA-melanin isolated from *Yarrowia lipolytica*. Very recently, a nitrate reductase enzyme purified from *F. oxysporum* was also successfully used for the production of biologically active silver nanoparticles (Gholami-Shabani et al., 2014).

Increasing incidence of microbial resistance to clinically approved classes of antibiotics and the continuing emphasis on health care costs have made a strong push towards the development of new and effective antimicrobial agents (Goffeau, 2008). The remarkable antimicrobial activity of silver nanoparticles has been well-demonstrated against a broad spectrum of Gram-positive and Gram-negative bacteria including multi drug resistant human pathogens (Ingle et al., 2008). Silver nanoparticles, biologically synthesized from *Cryphonectria* sp., showed potent antibacterial activity against multidrug resistant *Staphylococcus aureus*, *Escherichia coli*, *Salmonella typhi*, and *Candida albicans* and also significantly enhanced the bactericidal activity of standard antibiotics (Dar et al., 2013). More recently, Mishra et al. (2014) have reported that gold nanoparticles derived from *Trichoderma* sp. showed their potential as a new generation antimicrobial agents. In addition, gold nanoparticles synthesized from *Penicillium brevicompactum* have also displayed *in vitro* cytotoxic activity against mouse mayo blast cancer cells (C_2C_{12}) (Mishra et al., 2011).

At this juncture, as a part of our continuing search to identify microorganisms with the potential to synthesize nanoparticles with amazing biological properties, interest has spurred on soil microbes residing in forest areas. To the best of our knowledge, only a few soil fungi have been reported to synthesize nanoparticles (Gade et al., 2008; Jain et al., 2011; Li et al., 2012; Muhsin and Hachim, 2014; Salunkhe et al., 2011). Indeed, soil fungi are relatively unexplored as a potential resource of novel bio-reductants for the extracellular synthesis of silver and gold nanoparticles. Even though a large number of studies have focused on the antimicrobial activity of biosynthesized nanoparticles, the search for new nanoparticles with distinct physicochemical and biological properties remains at the forefront of current nanobiotechnological research (Elbeshehy et al., 2015). Therefore, the present study was aimed to isolate soil fungi from the forest areas of Kolli and Yer-

caud Hills, South India with the ultimate objective of producing antimicrobial nanoparticles.

2. Material and methods

2.1. Chemicals

Silver nitrate and gold chloride were purchased from HiMedia Laboratories Pvt., Ltd, Mumbai, India. The human pathogenic microorganisms were obtained from the Microbial Type Culture Collection (MTCC) centre, Chandigarh, India.

2.2. Study area

In this present study, a total of 20 soil samples were collected from the different forest areas of Kolli Hills, Namakkal (11.42N and 78.57416E) and 18 soil samples were collected from Yercaud Hills, Yercaud (12.309N and 78.343E) in the Province of Tamil Nadu, South India. The soil samples were collected in sterile polypropylene bags and were brought to the laboratory for isolation of fungi.

2.3. Isolation of soil fungi

Ten grams of soil samples were suspended in 100 mL of sterile water and these suspensions were considered as 10^{-1} dilution. Serial dilutions were done and 10^{-4} , 10^{-5} , and 10^{-6} dilutions were used for obtaining pure culture using potato dextrose agar (PDA) medium. The medium was also amended with chloramphenicol (25 $\mu\text{g}/\text{mL}$) to minimize bacterial contamination. Following serial dilution, the Petri plates were incubated at room temperature ($28 \pm 2^\circ\text{C}$) for 7 days. The Petri dishes were observed at regular intervals from the second day onwards for the fungal growth. Individual colonies of fungi were isolated and maintained on PDA slants.

2.4. Identification of soil fungi

The morphological identification of fungal isolates was determined by bright field microscopy observations of lacto phenol cotton blue stained fungal specimen at $40\times$ magnification. The isolated fungal strains were identified on the basis of spore morphology down to genus level by standard mycological monographs (Nag Raj, 1993; Sutton, 1980). In this present study, a total of 65 different fungal isolates were obtained and identified.

2.5. Screening of soil fungi for mycogenic synthesis of silver and gold nanoparticles

All the 65 fungi were screened for the biogenic synthesis of both silver and gold nanoparticles. To prepare the biomass, the fungi were grown aerobically in potato dextrose broth (PDB) and were incubated at 27°C for 7 days. After incubation, the profusely grown fungal mat was washed extensively using sterile double distilled water to remove the traces of medium components. Typically, 10 g (wet weight) of fungal mat was brought in contact with 100 mL sterile double distilled water in an Erlen Meyer flask and was kept under shaker condition (120 rpm) for 48 h at 27°C . Then, the mycelial free filtrate was obtained by passing it through Whatman filter paper No. 1. The filtrate was reacted with known quantity of silver nitrate to yield an overall Ag^+ ion concentration of 10^{-3} M and the reaction was carried out in dark at room temperature. Concurrently, the mycelial free extract and silver nitrate solution were maintained as controls and the change in color was observed up to 48 h. For gold nanoparticle synthesis, gold chloride was added in the place of silver nitrate and the same conditions were provided (Fayaz et al., 2010).

2.5.1. 18S rDNA sequencing and phylogenetic analysis

Among the 65 soil fungi tested, the isolate Bios PTK 6 synthesized extremely stable silver and gold nanoparticles within 12 h; hence, it was further chosen for molecular identification. The fungal genomic DNA was isolated by CTAB extraction method according to the procedure of Stirling (2003) and PCR amplification was done using ITS 1 and ITS 2 primers as described by White et al. (1990). The part of ribosomal RNA spanning the 3' end of the 18S rDNA, the internal transcriber spacer, the 5.8S rDNA and a part of the 5' end of 28S rDNA was amplified. The purified PCR products were sequenced using ABI 3730xl Genetic Analyzer by BigDye terminator method (Applied Biosystems, USA). The 18S rDNA sequence of this fungus was aligned with reference sequences showing sequence homology from the NCBI database using multiple sequence alignment programme. Phylogenetic tree was constructed by maximum parsimony analysis method using Close Neighbor-Interchange (CNI) algorithm. All positions containing gaps and missing data were eliminated from the dataset (complete deletion option). The tree was rooted using *Amanita muscaria* (accession number EU236711.1) as an out group. The stability of trees obtained from the above cluster analysis was assessed using bootstrap programme in sets of 1000 resamplings by MEGA4 (Tamura et al., 2007).

2.5.2. Optimization of silver and gold nanoparticles

Different concentrations of fungal biomass such as 10, 20, and 30 g/100 mL were optimized for the better synthesis of silver nanoparticles. Different concentrations of silver nitrate ranging from 1 to 10 mM were added to the mycelial free extract and incubated at different pH values (1–10) and temperatures (30–80 °C) for various time periods up to 48 h. The pH was adjusted using 0.1 N HCl or 0.1 N NaOH solutions and the absorbance of the resulting solution (after color change) was measured spectrophotometrically. Similarly, the mycosynthesis of gold nanoparticles was also optimized using same reaction parameters but with different substrate (gold chloride). In addition, the stability of as-synthesized silver and gold nanoparticles was also monitored (Krishnaraj et al., 2012; Mishra et al., 2014).

2.6. Characterization of mycosynthesized silver and gold nanoparticles

Preliminary characterization of silver and gold nanoparticles was done through visual observation for change in color. Time-dependent formation of silver and gold nanoparticles was observed using UV–vis. spectrophotometer. The mycosynthesized silver nanoparticles was confirmed by sampling the reaction mixture at regular intervals and the absorption spectra was scanned at the wavelength of 300–700 nm in Hitachi-U 2900 spectrophotometer. For electron microscopic studies, 10 μ L of mycosynthesized silver nanoparticles was drop coated on carbon grid and the images of nanoparticles were studied using high resolution-transmission electron microscopy (HR-TEM) assisted with energy dispersive X-ray spectroscopy (EDX) (FEI Tecnai); also, selected area electron diffraction (SAED) pattern was analyzed. For fourier transform infrared spectroscopic (FTIR) analysis, the mycosynthesized silver nanoparticles were freeze-dried and then diluted with potassium bromide in the ratio of 1:100. The FTIR spectrum was recorded using Shimadzu IR Prestige-21 FTIR instrument with a diffuse reflectance mode (DRS-8000) attachment. All measurements were carried out in the range of 400–4000 cm^{-1} at a resolution of 4 cm^{-1} . For X-ray diffraction (XRD) studies, dried nanoparticles were coated on XRD grid and the spectra were recorded by using Philips PW1830 X-ray generator operated at a voltage of 40 kV and a current of 30 mA with Cu K α 1 radiation. Further, the concentration of mycosynthesized silver nanoparticles was determined using inductively coupled

plasma-atomic emission spectrometer (ICP-AES) (ARCOS, Spectro, Germany). An electrokinetic measurement of silver nanoparticles as a function of pH in 1×10^{-3} mol dm $^{-3}$ aqueous solution was evaluated using zetasizer (Malvern Instruments, Worcestershire, United Kingdom). Aforementioned techniques were also adopted for the characterization of gold nanoparticles.

2.7. Antimicrobial activity of myco-derived silver and gold nanoparticles

2.7.1. Well diffusion assay

In this study, seven different human pathogenic bacteria *E. coli* ATCC 8739, *Pseudomonas aeruginosa* ATCC 27853, *Klebsiella pneumoniae* ATCC 2719 (Gram-negative), *Enterococcus faecalis* ATCC 29212, *Bacillus subtilis* ATCC 6633, methicillin-resistant *S. aureus* (MRSA), *S. aureus* ATCC 29736 (Gram-positive) and a yeast pathogen, *C. albicans* ATCC 2091 were used as test organisms and grown in Mueller-Hinton agar (MHA) medium. Well diffusion assay was performed to determine the antimicrobial activity of mycosynthesized silver and gold nanoparticles. Twenty five micro litres of silver and gold nanoparticles were loaded separately into each well of the Petri plates. Mycelial free extract was used to compare the antimicrobial activity of synthesized nanoparticles; also streptomycin (1 mg/mL) was used as a positive control for bacterial strains while fluconazole (1 mg/mL) was adopted in case of yeast pathogen. After inoculation, the plates were incubated at 37 °C for 24 h and the zone of inhibition (ZOI) was measured in terms of millimetre. These assays were carried out in triplicate.

2.7.2. Minimum inhibitory concentration of silver nanoparticles

The minimum inhibitory concentration (MIC) of silver nanoparticles was determined by broth microdilution method given by the Clinical Laboratory Standards Institute (CLSI) using 96-well microtitre plates (Singh et al., 2013). Two-fold serial dilutions of known concentrations of silver nanoparticles (100, 50, 25, 12.5, 6.25, 3.12, 1.5, 0.78, 0.39, and 0.19 μ g/mL) and antibiotics were made using Mueller-Hinton broth with appropriate control. To each well, 5 μ L inoculum ($\approx 5 \times 10^5$ CFU/mL) was added and the plates were incubated at 37 °C for 20 h. The lowest concentration which completely inhibited the growth of microbes was recorded as MIC. From the above assay, a loopful of inoculum was taken from each well showing no visual growth after incubation and spotted onto MHA plates to validate the MIC assay. All the experiments were performed in triplicate.

2.8. Mechanism of action of silver nanoparticles on microbial cells

2.8.1. Measurement of cellular leakage

2.8.1.1. Protein leakage assay. The protein leakage assay was carried out using the method of Kim et al. (2011). The bacterial cells were treated with known concentration of silver nanoparticles for about 3 and 6 h and then centrifuged at 6000 rpm for 15 min. For each sample, 200 μ L of the supernatant was mixed with 800 μ L of Bradford reagent and then incubated for 10 min. The optical density was measured at 595 nm using Hitachi-U 2900 spectrophotometer. Bovine serum albumin (BSA) was used as a standard protein and the experiments were done in triplicate.

2.8.1.2. Nucleic acid leakage assay. The nucleic acid leakage study was performed following the procedure of Álvarez-Ordóñez et al. (2013) with little modifications. Briefly, aliquots of 3 mL of bacterial cultures were exposed to silver nanoparticles at different time intervals such as 3 and 6 h. Then, it was filtered using a 25 mm diameter, 0.2 μ m pore size Millex-GS syringe filter (Millex-GS, Millipore, Madrid, Spain). The presence of nucleic acids in the filtrate was checked by measuring the absorbance at 260 nm using UV–vis.

Table 1

Screening of soil fungi for the biogenic synthesis of silver and gold nanoparticles.

(a) Fungi isolated from the soil samples collected from Kolli Hills				
Sl. no.	Strain no.	Name of the soil fungus	Silver nanoparticles synthesis	Gold nanoparticles synthesis
1	BIOS PTK1	<i>Aspergillus fumigatus</i>	+	+
2	BIOS PTK2	<i>Aspergillus niger</i>	+	+
3	BIOS PTK3	<i>Aspergillus niger</i>	+	–
4	BIOS PTK4	<i>Penicillium</i> sp.	–	–
5	BIOS PTK5	<i>Aspergillus terreus</i>	+++	+++
6	BIOS PTK6	<i>Aspergillus terreus</i>	+++	+++
7	BIOS PTK7	<i>Aspergillus niger</i>	+++	+++
8	BIOS PTK8	<i>Aspergillus</i> sp.	–	–
9	BIOS PTK9	<i>Aspergillus fumigatus</i>	–	–
10	BIOS PTK10	<i>Aspergillus flavus</i>	+	+
11	BIOS PTK11	<i>Trichoderma</i> sp.	++	++
12	BIOS PTK12	<i>Fusarium</i> sp.	++	–
13	BIOS PTK13	<i>Aspergillus fumigatus</i>	+	+
14	BIOS PTK14	<i>Aspergillus</i> sp.	–	–
15	BIOS PTK15	<i>Penicillium</i> sp.	–	–
16	BIOS PTK16	<i>Penicillium</i> sp.	–	–
17	BIOS PTK17	<i>Fusarium</i> sp.	–	–
18	BIOS PTK18	<i>Trichoderma</i> sp.	+	+
19	BIOS PTK19	<i>Mucor</i> sp.	–	–
20	BIOS PTK20	<i>Verticillium</i> sp.	–	–
21	BIOS PTK21	<i>Aspergillus flavus</i>	–	–
22	BIOS PTK22	<i>Aspergillus flavus</i>	+	+
23	BIOS PTK23	<i>Aspergillus flavus</i>	–	–
24	BIOS PTK24	<i>Aspergillus niger</i>	+	–
25	BIOS PTK25	<i>Aspergillus flavus</i>	+	+
26	BIOS PTK26	<i>Aspergillus flavus</i>	–	–
27	BIOS PTK27	<i>Aspergillus</i> sp.	+	–
28	BIOS PTK28	<i>Penicillium</i> sp.	+	–
29	BIOS PTK29	<i>Aspergillus fumigatus</i>	–	–
30	BIOS PTK30	<i>Aspergillus flavus</i>	+	+
31	BIOS PTK31	<i>Aspergillus fumigatus</i>	++	–
32	BIOS PTK32	<i>Fusarium</i> sp.	+	–
33	BIOS PTK33	<i>Aspergillus fumigatus</i>	+	+
34	BIOS PTK34	<i>Aspergillus flavus</i>	–	–
35	BIOS PTK35	<i>Aspergillus</i> sp.	–	–
36	BIOS PTK36	<i>Aspergillus</i> sp.	+	–
37	BIOS PTK37	<i>Aspergillus fumigatus</i>	+	+
38	BIOS PTK38	<i>Fusarium</i> sp.	+	–
39	BIOS PTK39	<i>Aspergillus niger</i>	+	+
(b) Fungi isolated from the soil samples collected from Yercaud Hills				
Sl. no.	Strain no.	Name of the fungus	Silver nanoparticles synthesis	Gold nanoparticles synthesis
1	BIOS PTK40	<i>Aspergillus</i> sp.	+	–
2	BIOS PTK41	<i>Trichoderma</i> sp.	+	–
3	BIOS PTK42	<i>Fusarium oxysporum</i>	+++	+++
4	BIOS PTK43	<i>Penicillium</i> sp.	–	–
5	BIOS PTK44	<i>Aspergillus</i> sp.	–	–
6	BIOS PTK45	<i>Aspergillus nidulans</i>	–	–
7	BIOS PTK46	<i>Trichoderma</i> sp.	++	+
8	BIOS PTK47	<i>Aspergillus</i> sp.	+	–
9	BIOS PTK48	<i>Mucor</i> sp.	–	–
10	BIOS PTK49	<i>Aspergillus</i> sp.	–	–
11	BIOS PTK50	<i>Aspergillus</i> sp.	+	–
12	BIOS PTK51	<i>Trichoderma</i> sp.	++	–
13	BIOS PTK52	<i>Fusarium</i> sp.	+	–
14	BIOS PTK53	<i>Aspergillus</i> sp.	–	–
15	BIOS PTK54	<i>Aspergillus</i> sp.	+	–
16	BIOS PTK55	<i>Aspergillus</i> sp.	–	–
17	BIOS PTK56	<i>Aspergillus niger</i>	++	++
18	BIOS PTK57	<i>Paecilomyces</i> sp.	–	–
19	BIOS PTK58	<i>Aspergillus fumigatus</i>	–	–
20	BIOS PTK59	<i>Aspergillus</i> sp.	+	–
21	BIOS PTK60	<i>Aspergillus flavus</i>	+	+
22	BIOS PTK61	<i>Penicillium</i> sp.	–	–
23	BIOS PTK62	<i>Fusarium</i> sp.	+	+
24	BIOS PTK63	<i>Fusarium</i> sp.	+	–
25	BIOS PTK64	<i>Aspergillus fumigatus</i>	–	–
26	BIOS PTK65	<i>Paecilomyces</i> sp.	–	–

(–) Negative for nanoparticles synthesis; (+) positive for nanoparticles synthesis; (++) nanoparticles synthesized below 12 h; (+++) nanoparticles synthesized below 12 h & stable over 2 months.

spectrophotometer. This experiment was repeated thrice for reproducibility.

2.8.2. SEM analysis

To understand the mode of action of silver nanoparticles on human pathogenic microorganisms, SEM analysis was carried out using the method of Ma et al. (2011) with slight modifications. After treatment with silver nanoparticles for 6 h, 1.0 mL each of test pathogen was withdrawn, centrifuged, and washed three times with phosphate buffered saline (PBS). The microbial cells were then fixed with 2.5% glutaraldehyde for 4 h and washed twice with PBS. After fixation, the cells were concentrated by centrifugation at 4000 rpm for 10 min, followed by washing twice with PBS buffer. Subsequently, the cells were gradually dehydrated with increasing concentrations of ethanol 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, and 100% for 5 min each. After drying, the cells were observed under SEM.

2.9. Statistical analysis

Data were analysed using one-way analysis of variance (ANOVA) followed by Sidak's post hoc analysis for leakage experiments and p values < 0.05 were considered significant. Statistical analyses were performed using GraphPad Prism 6 version software. The values are the mean of three independent experiments $\pm$ SEM.

3. Results and discussion

3.1. Screening of soil fungi for biogenic synthesis of silver and gold nanoparticles

Soil fungi are regarded as one of the versatile sources for the extracellular synthesis of metal nanoparticles. Forest environment is the largest reservoir consisting biological as well as chemical diversity. Hence, research focus on forest environment has been gaining momentum in the recent past. Even so, forest areas have not yet fully explored and there is tremendous scope to identify novel microorganisms having interesting biological activities. Along these lines, the present study was aimed to isolate soil fungi from the Kolli and Yercaud Hills, South India with the ultimate objective of producing antimicrobial nanoparticles. Altogether, 38 soil samples were collected from Kolli and Yercaud Hills. Among them, a total of 39 fungi were isolated from the soil samples of Kolli Hills (Table 1a). From the Yercaud Hills samples, 26 fungal organisms were isolated (Table 1b). Totally, 65 fungi were isolated from 38 soil samples using PDA medium. All the 65 fungi were screened for their ability to synthesize both silver and gold nanoparticles. Among them, 20 isolates showed positive response for both the metal nanoparticle synthesis; of these, 6 fungal species produced both silver and gold nanoparticles within 12 h (Table 1a and b). On the other hand, 38 isolates synthesized only the silver nanoparticles whereas 20 isolates produced gold nanoparticles alone (Table 1a and b). Among 41 species of *Aspergillus* screened, 25 species have produced either silver or gold nanoparticles extracellularly. Out of the 8 strains of *Fusarium* tested, 7 responded well for the mycosynthesis of silver and/or gold nanoparticles. Surprisingly, all the five isolates of *Trichoderma* produced either silver or gold nanoparticles. Of the 6 species screened, only one species of *Penicillium* has synthesized silver nanoparticles but it failed to synthesize gold nanoparticles. On the contrary, the members of *Verticillium*, *Mucor*, and *Paecilomyces* did not give positive signal for any of the metal nanoparticle synthesis (Table 1a and b).

In this study, only 4 isolates have synthesized extremely stable nanoparticles within 12 h (Table 1a and b). However, the isolate designated as Bios PTK 6 (Fig. S1) produced both the metal nanoparticles within 12 h with much greater stability. Further, species level

identification was also performed using molecular characterization for Bios PTK 6. Interestingly, the isolate Bios PTK 6 showed clustering with different species of *Aspergillus* with 100% bootstrap value in phylogram (Fig. S2). The complete molecular studies together with classical taxonomy results strongly supported the identification of isolated strain as *Aspergillus terreus*. Furthermore, this strain was deposited in the Microbial Type Culture Collection (MTCC) centre, Chandigarh, India with the reference number MTCC12235. In this study, based on the particle stability and faster rate of synthesis, *Aspergillus* spp. were recognized as potential strains for the extracellular synthesis of metal nanoparticles. In a similar fashion, many researchers have also utilized *Aspergillus* spp. as promising candidates for either silver or gold nanoparticle synthesis (Gade et al., 2008; Gupta and Bector, 2013; Jain et al., 2011; Li et al., 2012; Rodrigues et al., 2013).

3.2. Optimization of silver and gold nanoparticles using *A. terreus*

Controlling size, shape, and stability of nanoparticles is important for effective industrial applications (Mishra et al., 2011). Using *A. terreus*, different parameters such as biomass concentration, concentration of metal ions, pH, temperature, and time were optimized to control size, shape, and particle stability. Among the different concentrations of fungal biomass tested, 10 g/100 mL supported the better synthesis of both silver and gold nanoparticles whereas other tested concentrations (20 g/100 mL and 30 g/100 mL) did not favour the synthesis due to the excess amount of reducing agents (Fig. S3a and b). Among the different molar concentrations of silver nitrate tested, 1 mM concentration greatly enabled the silver nanoparticle synthesis with good monodispersity (Fig. 1a). Even though other concentrations of silver nitrate (2–10 mM) supported the mycosynthesis, 1 mM concentration was chosen in this study because it has much less toxicity compared to other concentrations. In case of gold nanoparticle synthesis, 1 mM concentration of gold chloride favored the synthesis whereas other concentrations (2–10 mM) did not support the nanoparticle formation (Fig. 2a).

The pH of the reaction medium is greatly influenced the nanoparticle formation as well as stability (Mishra et al., 2011). For silver nanoparticle production by *A. terreus*, there was no color change at acidic pH (1–4); however, aggregation of nanoparticles was observed at pH 3 and 4. Brown color formation began at pH 5 and 6 and the intensity of brown color was increased with the increase in pH (Fig. 1b); nevertheless, monodispersed and stable silver nanoparticles were formed at pH 7. In case of gold nanoparticle synthesis, there was no color change at pH 1 and 2. Pinkish violet color formation was observed at pH 3 and the intensity of pinkish violet color was found to be increased with increasing pH from 3 to 10 (Fig. 2b). Here also, pH 7 strongly supported the stable formation of gold nanoparticles. In addition to pH, temperature also influences the nanoparticle synthesis and stability (Mishra et al., 2014). Among the different levels of temperatures tested, both low and high temperatures supported the mycosynthesis of both silver and gold nanoparticles. Although higher temperatures favored the formation of silver and gold nanoparticles, however, 30 °C was found to be optimum for the stable formation of both the metal nanoparticles.

In this current study, the color change from pale yellow to brown and pinkish violet was observed within 12 h after incubation with 1 mM concentration of silver nitrate and gold chloride, respectively. However, *Nigrospora sphaerica*, isolated from the soil samples of cultivated localities, had synthesized silver nanoparticles after 72 h incubation with 1 mM silver nitrate solution (Muhsin and Hachim, 2014). *Aspergillus flavus* NJP08, another soil fungus, isolated from the metal-rich regions of Rajasthan had also shown brown color after 72 h incubation (Jain et al., 2011). Therefore, it is apparent that the present report shows almost six times faster rate of sil-

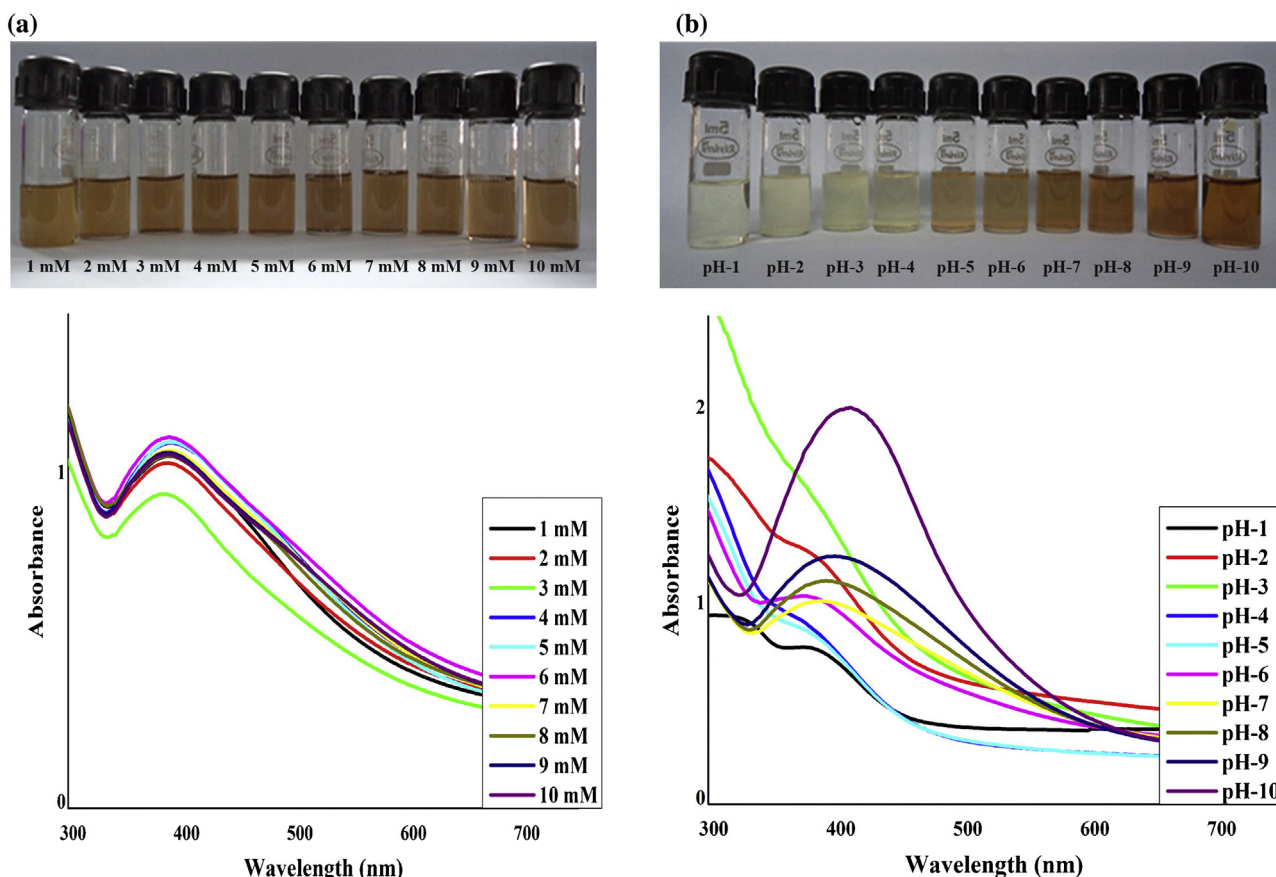


Fig. 1. Optimization of silver nanoparticles synthesized using *A. terreus* mycelial free filtrate. (a) Effect of different concentrations of silver nitrate on mycogenic synthesis of silver nanoparticles. All the tested concentrations of silver nitrate (1–10 mM) supported the nanoparticle synthesis as it can be seen from color change (inset) and UV–vis spectra. (b) Effect of pH on mycogenic synthesis of silver nanoparticles. There was no color change at pH 1–4; however, the aggregation of nanoparticles was observed at pH 3 and 4. Brown color formation began at pH 5 and 6 and the color intensity was found to be increased with increasing pH from 5 to 10 (inset). UV–vis absorption spectrum recorded at different pH conditions (after color change). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ver nanoparticle synthesis than the previous studies (Jain et al., 2011; Muhsin and Hachim, 2014). A recent report using new isolates of *Bacillus* spp. showed that silver nanoparticles were rapidly synthesized in less than 24 h of incubation (Elbeshehy et al., 2015), which was found to be two times lower rate of synthesis than our present study. In case of gold nanoparticles, our study shows two and four times faster rate of synthesis compared to earlier studies reported using *Aspergillus fumigatus* and *Aspergillus flavus*, respectively (Gupta and Bector, 2013).

3.3. Characterization of silver and gold nanoparticles

The very first indication for nanoparticle synthesis is color change. Brown and pinkish violet color formed in the optimized medium indicated the mycosynthesis of silver and gold nanoparticles, respectively. On the other hand, fungal biomass and substrate solution retained the original color. The color change was due to the excitation of surface plasmon vibrations, which is a characteristic feature of synthesized nanoparticles (Song et al., 2009). The absorption peaks recorded at 430 and 535 nm in the UV–vis spectra further confirmed the formation of silver and gold nanoparticles, respectively. HR-TEM image confirmed spherical shaped, well-dispersed silver nanoparticles (Fig. 3a) and the inset of Fig. 3a showed the SAED pattern. However, silver nanoparticles synthesized by *Bacillus* spp. were found to be triangular, hexagonal, and spherical (Elbeshehy et al., 2015). The particle size distribution analysis revealed an average size of 8–20 nm diameters by calculat-

ing 120 randomly selected silver nanoparticles in HR-TEM images (Fig. 3b). On the other hand, gold nanoparticles were of 10–20 nm in size with spherical shape; intriguingly, some particles were found to have anisotropic morphology with a size of 10–50 nm (Fig. 4a). SAED pattern of gold nanoparticles was also shown in the inset of Fig. 4a. The particle size distribution analysis of mycosynthesized gold nanoparticles showed an average diameter size of 15–30 nm (Fig. 4b).

The XRD analysis showed three distinct Bragg reflections corresponding to the (1 1 1), (2 0 0), and (2 2 0) orientations of the face-centered cubic (fcc) silver (Fig. 3c). The data obtained was matched well with the database of Joint Committee on Powder Diffraction Standards (JCPDS) file No. 04-0783, indicating the crystalline nature of silver nanoparticles. Similar to this, the XRD spectrum of gold nanoparticles also showed three clear diffraction peaks at planes corresponding to fcc gold (Fig. 4c) and was matched well with the JCPDS file No. 04-0784. The EDX spectrum also confirmed the presence of sharp peaks identical to silver (Fig. 3d) and gold (Fig. 4d) established in the optimized reaction medium, suggesting the successful synthesis of silver and gold nanoparticles. Zeta potential measurement confirmed the charge of the nanoparticles and it was found to be negative for both the metal nanoparticles (Figs. 3 e and 4 e). Much similar to our present report, Elbeshehy et al. (2015) have also shown a negative zeta potential for *Bacillus* spp. synthesized silver nanoparticles.

In addition, the concentration of silver as well as gold nanoparticles was also determined by ICP-AES analysis. Interestingly, A.

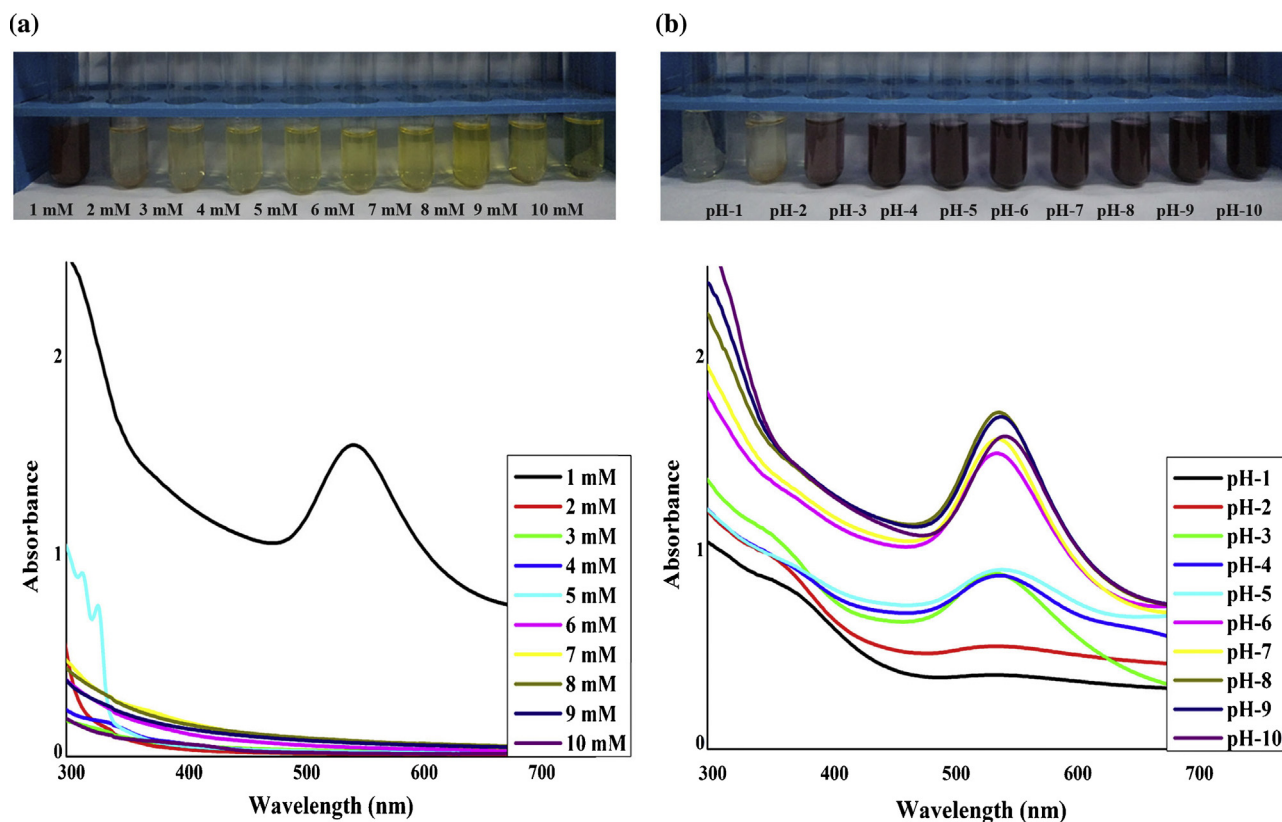


Fig. 2. Optimization of gold nanoparticles synthesized using *A. terreus* mycelial free filtrate. (a) Effect of different concentrations of gold chloride on mycogenic synthesis of gold nanoparticles. 1 mM concentration of gold chloride favored the synthesis whereas other concentrations (2–10 mM) did not support the nanoparticles formation as evidenced from color change (inset) and UV–vis spectra. (b) Effect of pH on mycogenic synthesis of gold nanoparticles. There was no color change at pH 1 and 2; pinkish violet color formation was observed at pH 3 and the color intensity was increased with increasing pH from 3 to 10 (inset). UV–vis absorption spectrum recorded at different pH conditions (after color change). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

terreus produced about 214 ppm of silver nanoparticles per litre of culture filtrate when 1 mM silver nitrate was challenged and for gold nanoparticles, it was 196 ppm. A common problem observed with nanoparticles is aggregation, which greatly decreases the surface area of the nanoparticles and, in turn, affects their physical, chemical, and biological properties (Sintubin et al., 2011). To assess the stability of silver and gold nanoparticles formed in the optimized medium, UV–vis. spectrophotometric study was carried out. In case of silver nanoparticles, there was no alteration in the peak at 430 nm even after 2 months of incubation period. Similarly, no change in the peak at 535 nm was also noticed for gold nanoparticles. However, the peaks for silver and gold nanoparticles got shifted during the course of third month but in both cases, there was no sign of aggregation of nanoparticles observed (Balakumaran, 2015). On the contrary, *Bacillus* spp. synthesized silver nanoparticles were found to be stable for up to 48 h (Elbeshehy et al., 2015).

FTIR analysis of the freeze-dried samples was carried out to identify the possible interactions between metal and bioactive molecules. The amide linkages between amino acid residues in proteins gave well known signatures in the infrared region of the electromagnetic spectrum (Jain et al., 2011). The FTIR spectrum of silver nanoparticles revealed two bands at 1605 and 1528 cm^{-1} , corresponding to the bending vibrations of the amide I and amide II bands of the proteins, respectively; their corresponding stretching vibrations were also observed at 3340 and 2924 cm^{-1} , respectively (Fig. 5a). It is well known that protein-nanoparticle interactions can occur either through free amine groups or cysteine residues in proteins and via the electrostatic attraction of negatively charged carboxylate groups in enzymes (Gole et al., 2001). Further, the two bands seen at 1366 and 1057 cm^{-1} can be assigned to the C–N

stretching vibrations of the aromatic and aliphatic amines, respectively (Vigneshwaran et al., 2007). The presence of the signature peaks of amino acids supported the presence of proteins in the mycelial free filtrate of *A. terreus*. This data was found to be consistent with our UV–vis. spectra of *A. terreus* mycelial free filtrate, which showed absorption peak at 280 nm, corresponding to the aromatic amino acids of proteins (Fig. S4). It is well-established that the absorbance peak arose at 280 nm was likely to the electronic excitations in tyrosine and tryptophan residues of the protein (Saifuddin et al., 2009; Xie et al., 2007). This indicates the presence of proteins in the mycelial free extract of *A. terreus*. Additionally, the EDX spectrum also showed signals for C, N, and O, which suggest the presence of proteins as capping agents on the surface of silver nanoparticles (Fig. 3d). Thus, it is obvious from these observations that the mycosynthesized nanoparticles are stabilized by the capping agent that is likely to be proteins present in the *A. terreus* mycelial free extract. At moderate concentrations of sucrose, glucose, and starch (30 g/L), *A. terreus* produced 0.76, 0.85, and 0.89 mg of extracellular protein per gram of dry cell weight, respectively (Han et al., 2010). Similarly, the FTIR spectrum of gold nanoparticles also showed similar bands as depicted in Fig. 5b. Interestingly, the FTIR results obtained in this study also revealed that the carbonyl groups from amino acid residues and peptides of proteins have strong affinity to bind with metals (Jain et al., 2011). However, the interaction between protein and nanoparticles is yet to be completely understood. On the basis of our observations, it is here hypothesized that the mycosynthesis of nanoparticles takes place via two steps: reduction and capping. At first, metal ions are reduced into respective metal nanoparticles, and secondly, capping of the synthesized nanoparticles occurs.

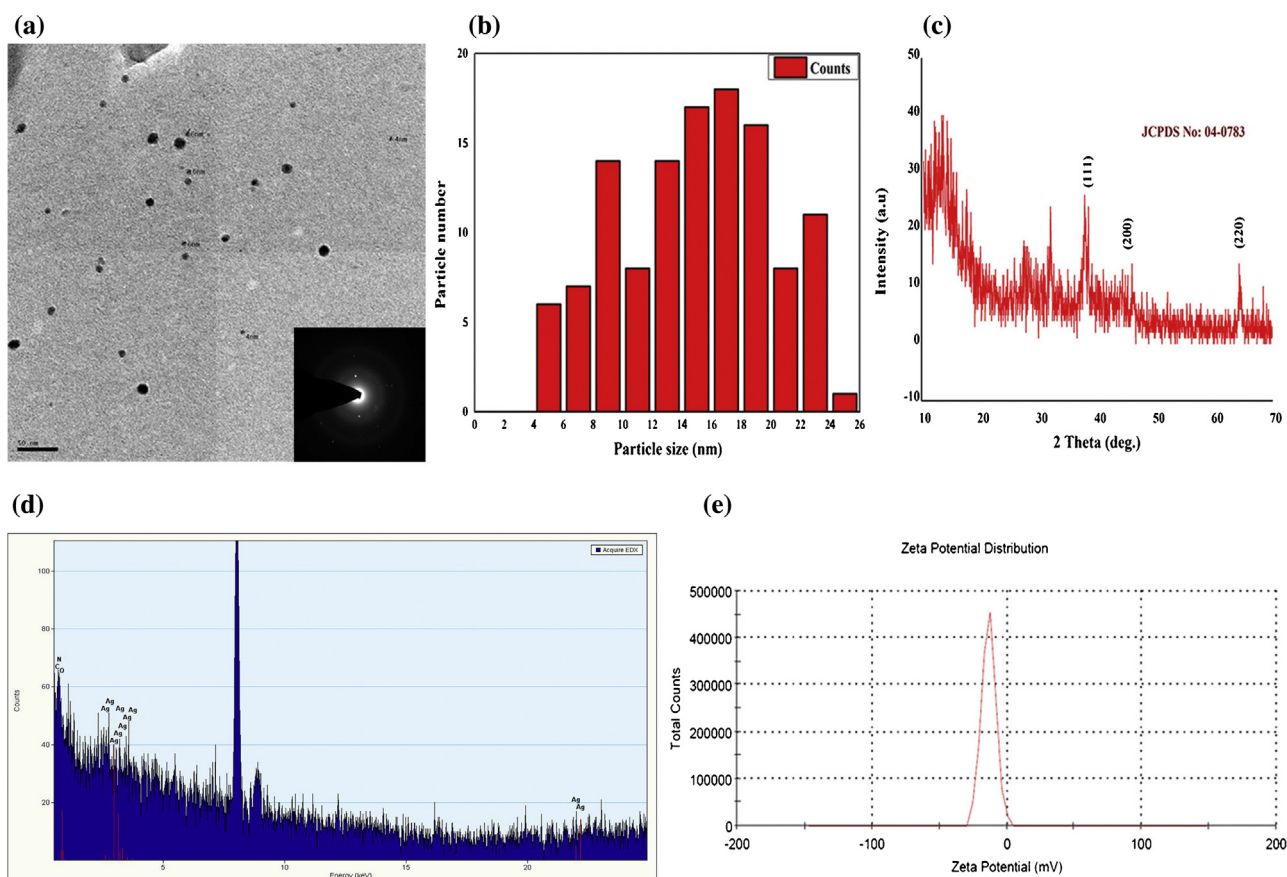


Fig. 3. Characterization of silver nanoparticles synthesized using *A. terreus* mycelial free filtrate. (a) High-resolution transmission electron microscopic image of spherical shaped silver nanoparticles. The size of the nanoparticles was 8–20 nm (scale bar = 50 nm) and the inset shows the SAED pattern. (b) Histogram analysis of the particle size distribution of silver nanoparticles. (c) X-ray diffraction pattern of *A. terreus* derived silver nanoparticles index at (1 1 1), (2 0 0) and (2 2 0), exhibiting the facets of crystalline silver. (d) Energy dispersive X-ray spectrum shows strong signal in the silver region and confirms the formation of silver nanoparticles from *A. terreus*. Other elemental signals were also recorded possibly due to elements from proteins present in the fungal extract. (e) Zeta potential measurement analysis of silver nanoparticles.

3.4. Antimicrobial activity of silver and gold nanoparticles synthesized from *A. terreus*

3.4.1. Well diffusion assay

In the preliminary screening assay, silver nanoparticles synthesized from *A. terreus* showed excellent antimicrobial activity. On the other hand, gold nanoparticles did not show any zone of inhibition (Balakumaran, 2015). Consequently, silver nanoparticles alone were used for further studies. The mycosynthesized silver nanoparticles showed superior antimicrobial activity against all the tested human pathogens and the data were presented in Fig. 6 and Table 2. Silver nanoparticles synthesized from *A. terreus* exhibited potent inhibitory activity against the pathogenic bacterium, *E. coli* with a clear inhibition zone of 22.47 mm at the concentration of

1 mg/mL. Similarly, silver nanoparticles significantly inhibited the growth of *E. faecalis* (19.47 mm), *P. aeruginosa* (17.13 mm), *B. subtilis* (16.43 mm), methicillin-resistant *S. aureus* (16.15 mm), *K. pneumoniae* (16.13 mm), *S. aureus* (14.27 mm), and *C. albicans* (18.13 mm) at a test concentration of 1 mg/mL. The zone of inhibition exhibited by silver nanoparticles was found to be higher than our previous study (Balakumaran et al., 2015) and slightly lower than those reported by Elbeshehy et al. (2015). The highest antimicrobial activity was recorded against Gram-negative bacteria and this result was in good agreement with the earlier studies (Balakumaran et al., 2015; Elbeshehy et al., 2015). Conversely, mycelial free extract did not show any inhibition zone against all the tested pathogens. Although silver nitrate showed a moderate activity, but due to its

Table 2
Antimicrobial activity of *A. terreus* derived silver nanoparticles against harmful human pathogens.

Test organism	Mycelial free extract (25 μ L)	Silver nitrate (1 mM)	Silver nanoparticles (1 mg/mL)	Streptomycin (1 mg/mL)
Mean zone of inhibition (mm)				
<i>E. coli</i>	–	12.67 $\pm$ 0.15	22.47 $\pm$ 0.06	14.27 $\pm$ 0.15
<i>P. aeruginosa</i>	–	15.67 $\pm$ 0.76	17.13 $\pm$ 0.15	13.33 $\pm$ 0.15
<i>K. pneumoniae</i>	–	12.23 $\pm$ 0.15	16.13 $\pm$ 0.06	5.67 $\pm$ 0.25
<i>E. faecalis</i>	–	16.47 $\pm$ 0.50	19.47 $\pm$ 0.06	17.53 $\pm$ 0.21
<i>B. subtilis</i>	–	12.73 $\pm$ 0.40	16.43 $\pm$ 0.21	24.37 $\pm$ 0.23
Methicillin-resistant <i>S. aureus</i>	–	8.83 $\pm$ 0.11	16.15 $\pm$ 0.07	5.97 $\pm$ 0.15
<i>S. aureus</i>	–	13.03 $\pm$ 0.06	14.27 $\pm$ 0.12	25.33 $\pm$ 0.15
<i>C. albicans</i> ^a	–	11.38 $\pm$ 0.17	18.13 $\pm$ 0.02	10.12 $\pm$ 0.10

Zone of inhibition was measured as millimeter $\pm$ standard deviation of at least three independent experiments; (–) no activity.

^a Fluconazole was used as a positive control for *C. albicans*.

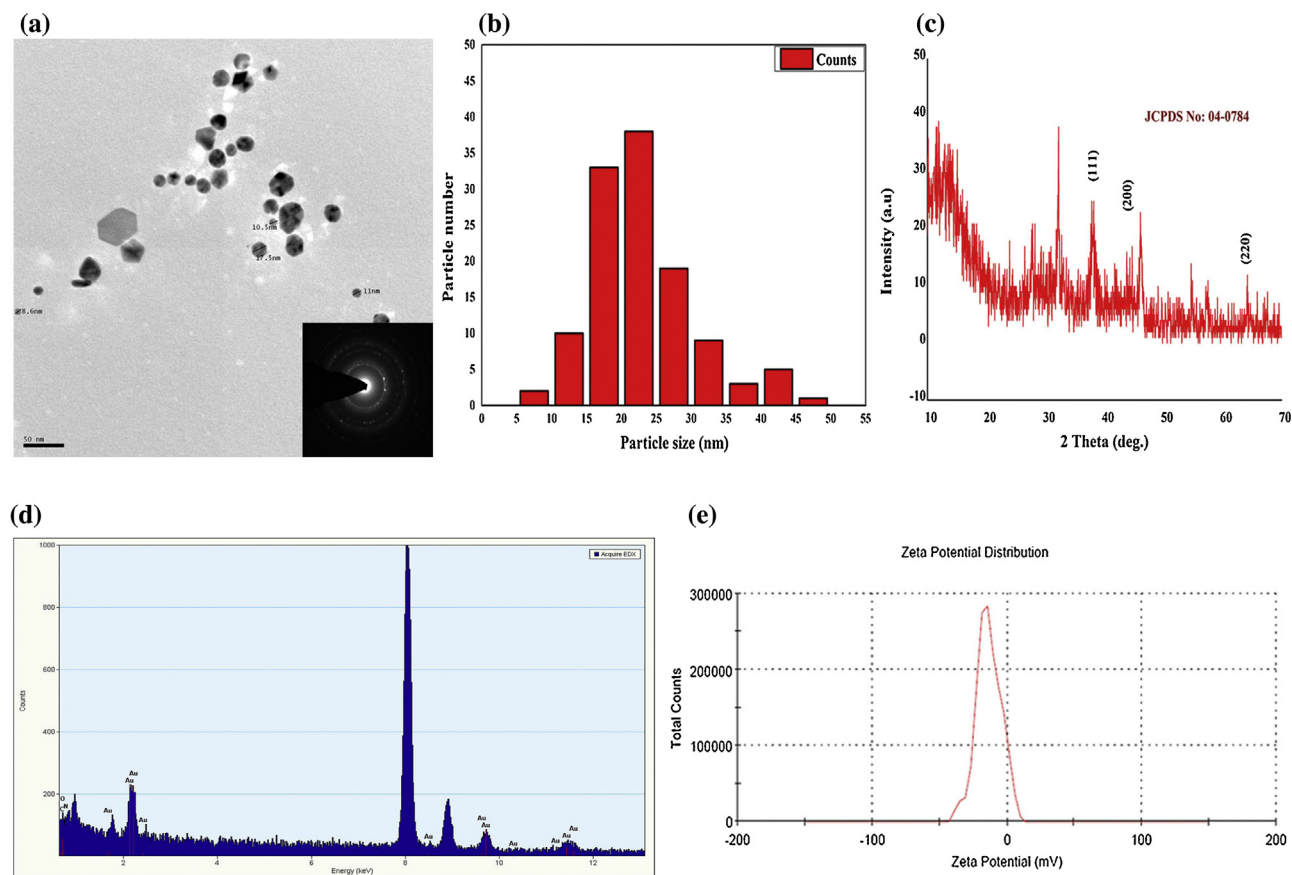


Fig. 4. Characterization of gold nanoparticles synthesized using *A. terreus* mycelial free filtrate. (a) High-resolution transmission electron microscopic image of gold nanoparticles showing anisotropic morphology. The size of the nanoparticles was 10–50 nm (scale bar = 50 nm); inset shows the SAED pattern. (b) Histogram analysis of the particle size distribution of gold nanoparticles. (c) X-ray diffraction pattern of *A. terreus* synthesized gold nanoparticles index at (1 1 1), (2 0 0) and (2 2 0), exhibiting the facets of crystalline gold. (d) Energy dispersive X-ray spectrum shows strong signal in the gold region and confirms the formation of gold nanoparticles from *A. terreus*. Other elemental signals were also recorded possibly due to elements from proteins present in the fungal extract. (e) Zeta potential measurement analysis of gold nanoparticles.

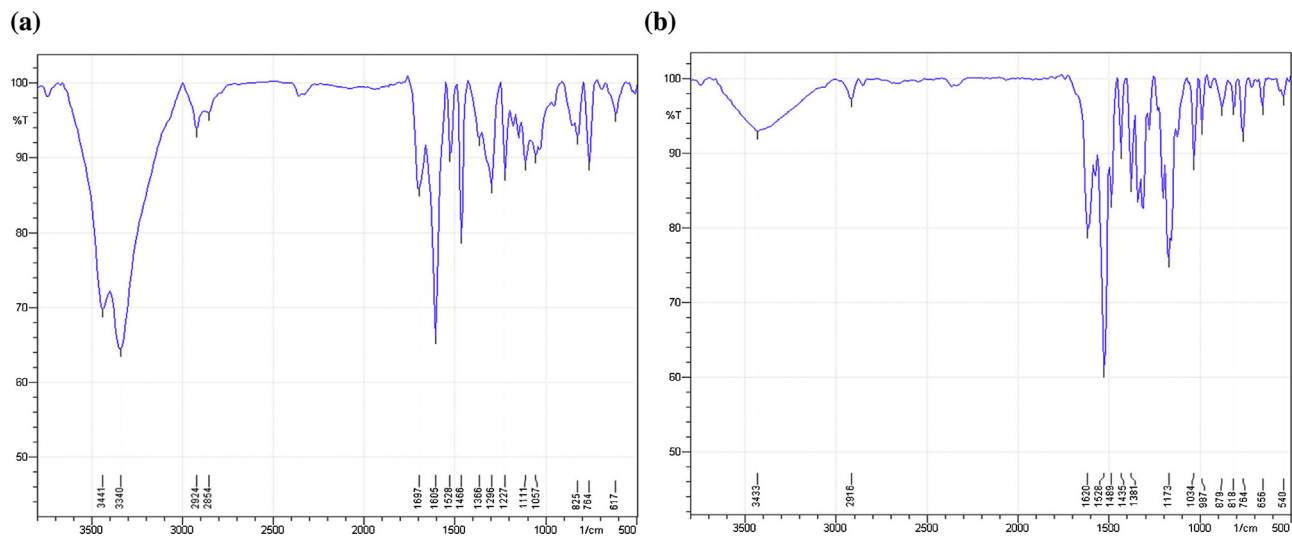


Fig. 5. Fourier transform infrared analyses of silver and gold nanoparticles synthesized using *A. terreus* mycelial free filtrate. (a) FTIR spectrum of silver nanoparticles. (b) FTIR spectrum of gold nanoparticles.

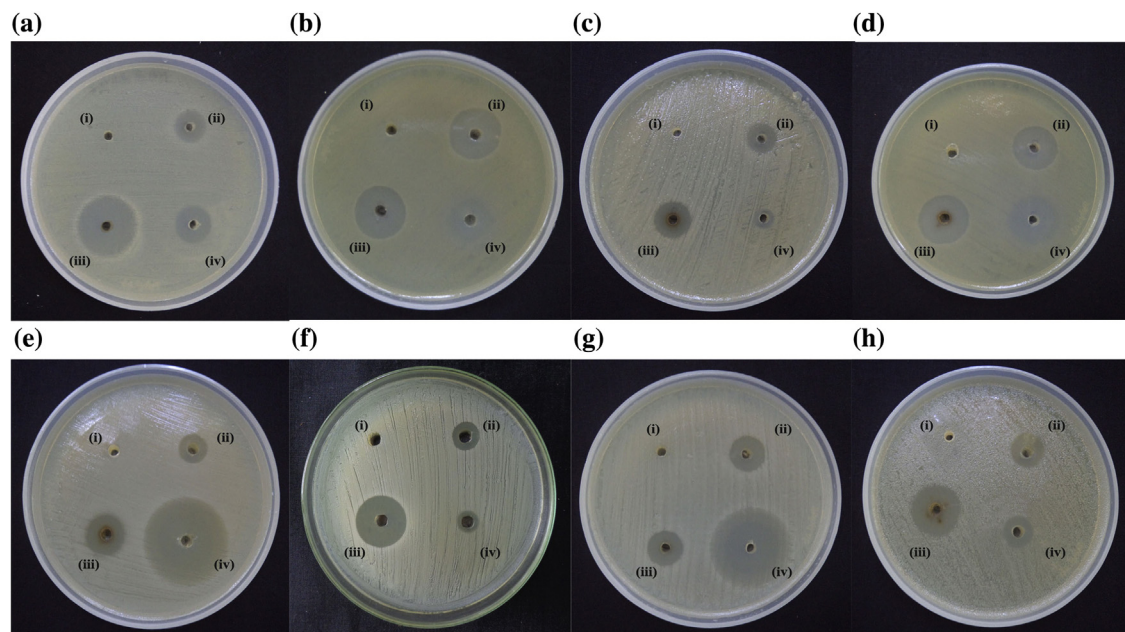


Fig. 6. Antimicrobial activity of *A. terreus* synthesized silver nanoparticles against human pathogenic microorganisms by well diffusion assay. (a) *E. coli*; (b) *P. aeruginosa*; (c) *K. pneumoniae*; (d) *E. faecalis*; (e) *B. subtilis*; (f) methicillin-resistant *S. aureus*; (g) *S. aureus*; (h) *C. albicans*. Each well was treated with (i) mycelial free extract (25 µL), (ii) silver nitrate (1 mM), (iii) silver nanoparticles (1 mg/mL) and (iv) streptomycin (1 mg/mL) for bacterial strains; fluconazole (1 mg/mL) in case of *C. albicans*.

strong cytotoxic nature on human cells, its effect was found to be insignificant.

3.4.2. Minimum inhibitory concentration of silver nanoparticles

MIC was recorded as the lowest concentration at which no visible growth of test pathogens was observed. In this present study, silver nanoparticles showed appreciable antimicrobial activity against *E. coli*, *E. faecalis*, methicillin-resistant *S. aureus*, and *C. albicans* exhibiting MIC of 3.125 µg/mL. Similarly, 6.25 µg/mL was found to be MIC for *P. aeruginosa*, *K. pneumoniae*, and *B. subtilis* while 12.5 µg/mL was proved as MIC against *S. aureus*. This result was found to be consistent with the studies of Elbeshehy et al. (2015). Additionally, the obtained results were validated using MHA plates which showed no organisms in it. Silver nanoparticles synthesized using *Bacillus licheniformis* showed lowest MIC and MBC values than

those synthesized using *Bacillus pumilus* and *Bacillus persicus* and this was likely due to the smaller particle size (Elbeshehy et al., 2015).

The higher activity of silver nanoparticles is probably due to the proteins or other biocompatible materials adsorbed on the surface of the synthesized nanoparticles. The adsorbed proteins and/or enzymes may augment the antimicrobial property of silver nanoparticles to some extent. Our FTIR results, as discussed above, reveal the presence and binding of proteins with silver nanoparticles, which evidently suggest that the released extracellular proteins assist the formation (reduction) and the possible stabilization (capping) of silver nanoparticles in the aqueous medium (Fig. 5a). Jain et al. (2011) have also supported this speculation. Therefore, it is apparent that the biomolecules especially proteins functionalized the synthesized nanoparticles and, in turn, could

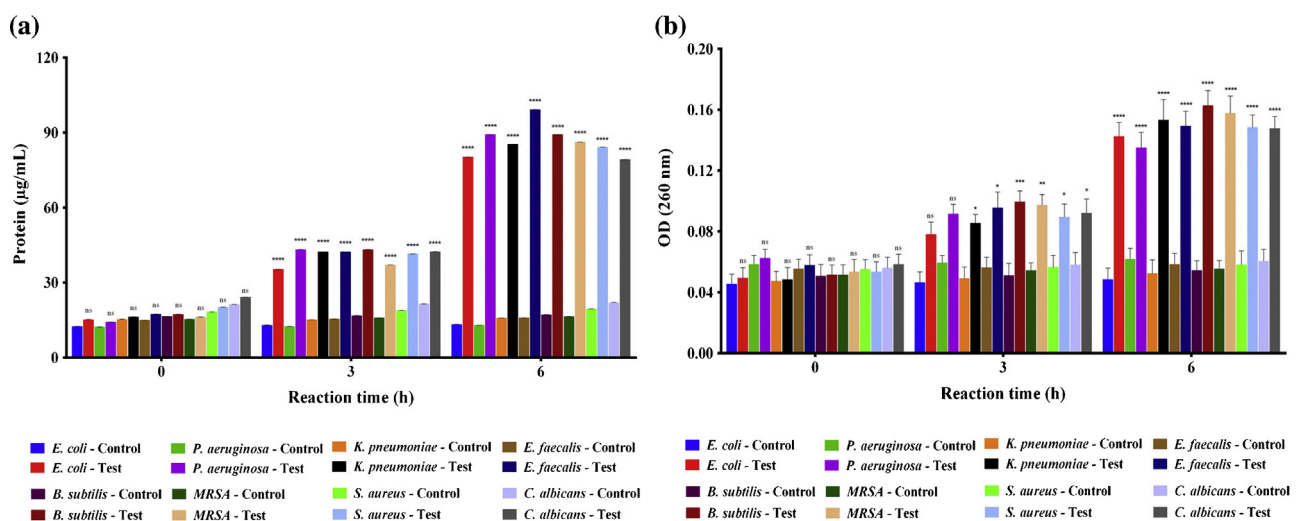


Fig. 7. The values are the mean of three independent experiments $\pm$ SEM. Statistical analysis was performed using one-way analysis of variance. Significance is reported as * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$ against the respective control values; ns—not significant. (a) Silver nanoparticles treated bacterial cells released highest amount of protein compared to control cells. (b) Silver nanoparticles treated cells showed highest amount of nucleic acid leakage than the control cells as it can be seen from increased OD values.

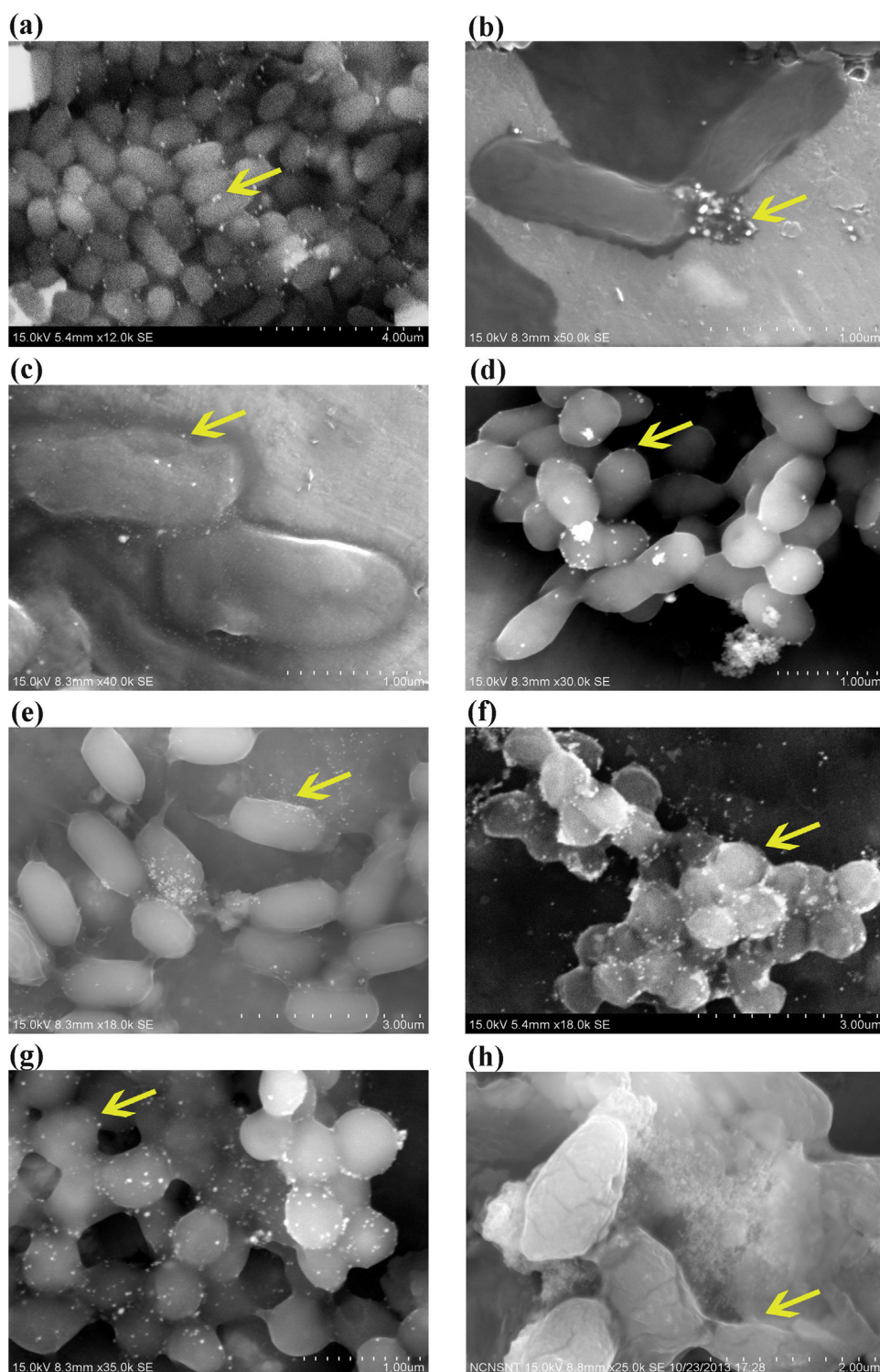


Fig. 8. Scanning electron micrographs of silver nanoparticles treated microbial cells at 6 h. (a) *E. coli*; (b) *P. aeruginosa*; (c) *K. pneumoniae*; (d) *E. faecalis*; (e) *B. subtilis*; (f) methicillin-resistant *S. aureus*; (g) *S. aureus*; (h) *C. albicans*. Arrow indicates the accumulation of nanoparticles over the cell membrane and the breakage of cell wall after treatment with silver nanoparticles.

increase the antimicrobial activity. As expected, because of their greater biocompatibility, the mycosynthesized silver nanoparticles have displayed pinnacle antimicrobial activity than the standard antibacterial drug, streptomycin (Fig. 6 and Table 2). It is also interesting to note that biomolecules such as proteins/peptides, carbohydrates, lipids, and nucleic acids functionalized nanoparticles have already been shown outstanding antibacterial activities

against several human pathogenic bacteria (Veerapandian and Yun, 2011). Thus, it is clear from these observations that the antimicrobial activity of silver nanoparticles shown here was a synergistic effect and this hypothesis was widely accepted by many researchers (Chaloupka et al., 2010).

3.5. Mechanism of action of silver nanoparticles on microbial cells

3.5.1. Measurement of cellular leakage

3.5.1.1. Protein leakage. Protein leakage from the membranes of bacterial cells treated with silver nanoparticles was almost the same as that from cells in the control group at the beginning of the experiment. However, at 3 h after treatment, the treated cells leaked nearly 35–43 $\mu\text{g/mL}$ protein while the untreated cells maintained the same level of leakage. At 6 h after incubation, the silver nanoparticles treated bacterial cells showed more than four times higher level of leakage than the normal control cells. Interestingly, *E. faecalis* cells showed highest amount of protein leakage (99 $\mu\text{g/mL}$) followed by *P. aeruginosa* (89 $\mu\text{g/mL}$) and *B. subtilis* (89 $\mu\text{g/mL}$) at 6 h after treatment with silver nanoparticles (Fig. 7a).

3.5.1.2. Nucleic acid leakage. The leakage of intracellular material after exposure to silver nanoparticles at different time interval was assessed by measuring the optical density at 260 nm (OD_{260}) of cell free filtrates. Initially, the nucleic acid leakage from nanoparticles treated cells and the control cells was almost the same. The bacterial cells exposed to silver nanoparticles treatment gave rise to the increased release of nucleic acids, which was more marked with *B. subtilis* followed by MRSA and *K. pneumoniae* at 3 and 6 h after treatment with silver nanoparticles (Fig. 7b). However, the control bacterial cells showed lesser OD values than the silver nanoparticles treated group.

3.5.2. SEM analysis

Further, SEM analysis was performed to understand the mode of action of silver nanoparticles on human pathogenic microorganisms. This study revealed the accumulation of silver nanoparticles over the cell wall, suggesting that nanoparticles entered through pores present on the cell membranes. In addition, cell wall breakage was also observed due to the action of silver nanoparticles (Fig. 8).

Although the bactericidal effects of silver nanoparticles have extensively been studied, the mechanism of action is not yet fully understood. However, fewer hypotheses have been proposed to give some insights on the antibacterial action of nano-sized silver particles. The bactericidal activity of silver nanoparticles is relied much on the size of the particles (Lok et al., 2007; Morones et al., 2005; Pal et al., 2007). Interestingly, in this study, *A. terreus* synthesized 8–20 nm sized silver nanoparticles. It is well-established that silver nanoparticles of less than 20 nm diameters easily get attached to the sulfur-containing proteins of bacterial cell membranes, thus leading to greater permeability of the membrane (Gogoi et al., 2006; Morones et al., 2005; Sondi and Salopek-Sondi, 2004). In this study, the ability of silver nanoparticles to damage the bacterial membrane was evaluated by measuring the release of intracellular contents. Silver nanoparticles treated bacterial cells showed considerably increased amount of protein leakage compared to the control group (Fig. 7a); this indicates that silver nanoparticles could increase permeability (Kim et al., 2011) and affect membrane transport (Sondi and Salopek-Sondi, 2004) due to the serious damage of cell membrane structure. This result is found to be consistent with the nucleic acid leakage assay, which showed increased OD values in silver nanoparticles treated bacterial cells than the control (Fig. 7b). The increase in OD_{260} indicates the leakage of intracellular nucleic acids and thus, reflects a loss in membrane integrity (Álvarez-Ordóñez and Prieto, 2010; Sampathkumar et al., 2003). The leakage of intracellular components also suggests that silver nanoparticles may cause formation of pores in the cytoplasmic membrane (Álvarez-Ordóñez et al., 2013). Altogether, these observations clearly suggest that silver nanoparticles have direct contact with bacteria as the membrane morphology was severely disturbed (Sondi and Salopek-Sondi, 2004).

Additionally, it has also been reported that silver nanoparticles have the ability to enter inside the bacterium and therefore they react with sulfur-containing proteins in the interior of the cell, as well as with phosphorus-containing compounds such as DNA (Morones et al., 2005). Sondi and Salopek-Sondi (2004) have observed silver nanoparticles accumulation in *E. coli* bacterial membrane. Similarly, our SEM studies revealed the accumulation of silver nanoparticles over the cell wall, suggesting that nanoparticles entered through pores present on the cell membrane; in addition, cell wall breakage was also observed (Fig. 8). This implies that silver nanoparticles could penetrate inside the bacteria and cause further damage by interacting with DNA (Morones et al., 2005; Sondi and Salopek-Sondi, 2004). From TEM and proteomics analyses, it has been suggested that silver nanoparticles could interact with bacterial membrane, causing structural changes, dissipation of proton motive force, leading to cell death (Lok et al., 2006). However, in our study, the antimicrobial activity exhibited by silver nanoparticles was found to be synergy, although the actual role of proteins in reduction, capping, and antibacterial activity has been poorly understood. Therefore, the mechanism of antibacterial action of silver nanoparticles is somewhat complex and depends on the bio-physico-chemical interactions of silver nanoparticles with the bacterial cell (Nel et al., 2009).

4. Conclusions

In conclusion, Kolli and Yercaud Hills were for the first time explored for the nanoparticle synthesis using soil fungi. This is the first study in which both silver and gold nanoparticles were synthesized using the mycelial free filtrate of *A. terreus*. The mycosynthesis protocol employed in this present study was found to be simple, rapid, cost-effective, and environmentally benign and thus, this extracellular synthesis method has huge potential to become developed into simple bioprocess system for sustainable production of nanoparticles at larger amount. However, the molecular interactions between nanoparticles and fungal proteins need to be critically analyzed to understand the exact mechanism behind the mycosynthesis. Moreover, the remarkable antimicrobial activity of silver nanoparticles, shown in this study, indicated that the biomolecules such as proteins have functionalized the nanoparticles and thereby, significantly enhanced the antibacterial activity. Therefore, the molecular mechanisms underlying the antibacterial activity of silver nanoparticles must be thoroughly studied using cutting edge techniques such as microarray, proteomic, and transcriptomic analyses and we are currently working toward this end.

Conflict of interest

None.

Acknowledgements

The authors thank the Ministry of Human Resource Development (MHRD), Government of India, New Delhi for supporting this work under Special Grants through National Centre for Nanoscience and Nanotechnology (NCNSNT), University of Madras, Chennai. We are grateful to the Director, CAS in Botany, School of Life Sciences, University of Madras for providing adequate laboratory facilities. The authors are thankful to the Director, NCNSNT, University of Madras for SEM, HR-TEM and EDX analyses. The Head, Department of Nuclear Physics, University of Madras and the Head, SAIF, IIT-Madras are gratefully acknowledged for XRD and ICP-AES analysis, respectively.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.micres.2015.09.009>.

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